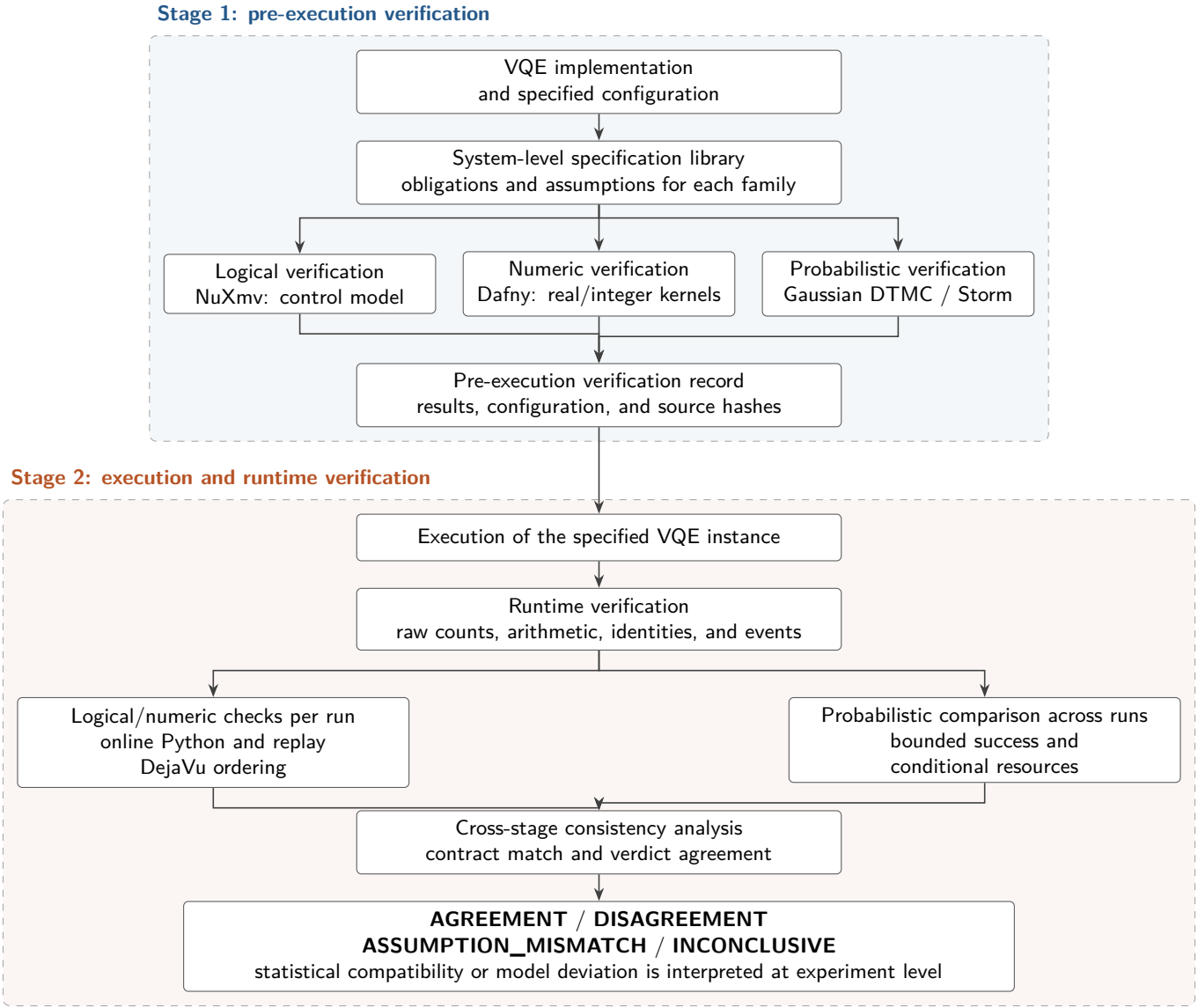
Stage 1: pre-execution verification
VQE implementation
and specified configuration
System-level specification library
obligations and assumptions for each family
Logical verification
NuXmv: control model
Numeric verification
Dafny: real/integer kernels
Probabilistic verification
Gaussian DTMC / Storm
Pre-execution verification record
results, configuration, and source hashes
Stage 2: execution and runtime verification
Execution of the specified VQE instance
Runtime verification
raw counts, arithmetic, identities, and events
Logical/numeric checks per run
online Python and replay
DejaVu ordering
Probabilistic comparison across runs
bounded success and
conditional resources
Cross-stage consistency analysis
contract match and verdict agreement
AGREEMENT / DISAGREEMENT
ASSUMPTION_MISMATCH / INCONCLUSIVE
statistical compatibility or model deviation is interpreted at experiment level